Notable Identities

Classical Identities in Quadratic Nodes, Rational Rotations and Linear Recurrences

Carolina Johnson (CJ)

November 2026

Abstract

The notable identities of elementary algebra, the *fórmulas notables* or *productos notables* of the Spanish-language curriculum, are placed at the steps they carry in a set of constructions: the quadratic node sequence $r_n = n^2 r_0$ and its odd intervals, the Geometric Matrix of four boundary variables, the Smith chart reflection coefficient, rational rotations with Pythagorean entries, the Fibonacci and Tribonacci recurrences, and the inscribed hexagon. Every step is shown. The difference of squares of Euclid's Elements II.5 carries the node intervals, the boundaries of the Geometric Matrix, the Smith chart ratio and the cubes, which are the differences $T(n)^2 - T(n-1)^2$ of squared triangular numbers. The identity $(x+y)^2 - (x-y)^2 = 4xy$ builds the rotation triple $(q^2 - p^2,\ 2pq,\ q^2 + p^2)$, and the nodes at which the odd interval is a square, here called Pythagorean nodes, are the triples of Pythagoras's rule. The two-square identity of Brahmagupta and Fibonacci composes rotations in the plane, and Euler's four-square identity carries the same step in three dimensions, where the quadruple $(3, 4, 12, 13)$ closes the space diagonal over the 3-4-5 triangle. The Sophie Germain identity is the difference of squares taken twice over the Gaussian integers at $e^{i\pi/4}$, and the difference of cubes together with Argand's identity splits the vertices of the inscribed hexagon of Euclid IV.15. Consecutive Fibonacci numbers give the rotation triples $(F_{n-1}F_{n+2},\ 2F_nF_{n+1},\ F_{2n+1})$, and the binomial powers, Cauchy's cube, Stevin's identity and its three-factor extension, Gauss's identity and the trinomial square and cube carry the power relations of the Lucas numbers and of the Tribonacci power sums. Each identity of the catalogue is placed, together with the sum of odd numbers in the arrangement of Nicomachus. The quadratic identities among them are propositions II.4 to II.10 of Euclid's Book II, stated there as relations among rectangles and squares on a cut line, and the Fibonacci equation is the reciprocal form of the cut in extreme and mean ratio of II.11 and VI.30. Two kinds of identity carry the steps. The difference of squares, the two-square and four-square identities, the cube factorizations and the Fibonacci relation $L_n^2 - 5F_n^2 = 4(-1)^n$ are norm relations in the split-complex numbers, the Gaussian integers, the Eisenstein integers, the integers of $\mathbb{Q}(\sqrt{5})$ and the quaternions, and the Tribonacci roots are the three conjugates of a cubic field. The binomial powers and the symmetric products are polynomial expansions, and they carry the relations among powers of these conjugates. The identities, their origins and this norm structure are classical; the paper adds the map that places each identity at a step of the constructions. The map makes each step auditable against a named classical identity and identifies the identities on which the exact computations depend. Each identity is also placed beside its classical source and the names it takes later in algebra, engineering and physics, so that one relation can be recognized under its several names.

1 Introduction

A notable identity is a polynomial equality that holds for all values of its variables and is used often enough to be learned by name. The Spanish-language curriculum teaches them as *fórmulas notables* or *productos notables*: the square of a binomial, the difference of squares, the cube of a binomial, the sum and difference of cubes, and a group of identities named after Legendre, Argand, Gauss, Cauchy, Stevin and Sophie Germain. Each is proved by expanding a product and collecting terms. They are notable because they appear in plain sight: they recur often enough in calculation that they are recognized on sight rather than recomputed.

Origin. The quadratic identities are older than their names. Euclid's Book II states them as relations among rectangles and squares on a line cut at one or two points: II.4 is the square of a sum, II.5 and II.6 are the difference of squares, II.7 is the square of a difference, and II.8 to II.10 are the identities the curriculum names after Legendre [6]. Heath's commentary gives the algebraic form of each proposition and reports the Greek attribution of the content of Book II to the Pythagoreans [6]. Neugebauer [23] and Høyrup [24] showed that Old Babylonian texts of the early second millennium BCE solve quadratic problems by cutting and rearranging rectangles in the same manner, and the tablet Plimpton 322 of the same period lists Pythagorean triples. Robson reads its rows as generated from a number x and its reciprocal, which in modern notation is $((x - 1/x)/2)^2 + 1 = ((x + 1/x)/2)^2$ [25], the identity $(x+y)^2 - (x-y)^2 = 4xy$ with $xy = 1$; this is a modern reconstruction, and the purpose and procedure of the tablet remain debated; Mansfield and Wildberger read the tablet as a table of exact sexagesimal trigonometry [36]. Unguru [26] argued that reading Book II as algebra projects later notation onto a geometry of magnitudes; the debate concerns interpretation, and the equivalences themselves are not in question. In the ninth century Thābit ibn Qurra proved the rules of al-Khwārizmī for quadratic equations from Euclid II.5 and II.6 [27], which joined the algebra to Euclid by name. Several curriculum names are later labels: the relations between roots and coefficients behind the three-factor extension of Stevin's identity are due to Viète [28] and Girard [29], and Binet's formula was known to de Moivre [30]. The eponymous names of the curriculum, Legendre, Argand, Gauss, Cauchy, Stevin and Sophie Germain, are kept as conventions; the historical origin of an identity is stated separately where it is established.

One relation, several names. The identities are taught, set aside once they become routine, and met again later under other names. The difference of squares returns as multiplication by a conjugate, the two-square identity as the multiplicativity of the complex modulus, Stevin's identity as Viète's relations between roots and coefficients, and Pythagoras's rule as the rational points of the unit circle. Each field teaches its own name, and the earlier one is no longer displayed. The Smith chart paper [3] calls this pattern structural continuity under nomenclature displacement: the same relation carried through successive vocabularies. Table 1 places each identity of the catalogue beside its classical source and the names it takes later.

This paper places these identities at the steps they carry in a set of constructions developed in earlier papers: the quadratic node sequence and its odd intervals [1], the Geometric Matrix [2], the partition ratio of Euclid II.5 and the Smith chart [3], the rational rotations of [1], the Fibonacci ratio [4] and the frequency composition of [5]. The constructions were developed from the arithmetic of the objects themselves, and the identities are the classical names of the relations that arithmetic produces. The paper therefore attributes each identity to its classical source and states where it enters each construction.

What is classical and what this paper adds. The identities are classical, and so is their origin:

Curriculum name	Classical source	Later names
Square of a sum and difference	Euclid II.4, II.7	binomial expansion; completing the square
Difference of squares	Euclid II.5, II.6; Babylonian	multiplication by a conjugate; rationalizing a denominator; split-complex norm; power balance of a lossless line
Legendre	Euclid II.8 to II.10	parallelogram law $\|a+b\|^2 + \|a-b\|^2 = 2(\|a\|^2 + \|b\|^2)$; polarization identity $4ab = (a+b)^2 - (a-b)^2$
Stevin and its extension	Viète, Girard	Vieta's relations; elementary symmetric polynomials
Cube, fourth power, Newton	Pascal's triangle, Newton	binomial theorem
Sum and difference of cubes, Argand	factorization of $a^3 \pm b^3$	Eisenstein norm; cyclotomic factors of $x^6 - 1$
Trinomial square and cube, Gauss	symmetric expansion	power sums of conjugate roots; determinant of a 3×3 circulant matrix
Brahmagupta–Fibonacci	Diophantus, Brahmagupta, Fibonacci	complex modulus $\|z\|^2 = z\bar{z}$ and $\|zw\| = \|z\|\|w\|$; composition of plane rotations; squared amplitude $\|\psi\|^2$
Euler four-square	Euler, Hamilton	quaternion norm; unit quaternions as rotations of space; spinor regularization
Sophie Germain	Sophie Germain	factorization over the Gaussian integers
Pythagoras's rule	Pythagoras, reported by Proclus; earlier triples on Plimpton 322	rational points of the unit circle; tangent half-angle substitution $t = \tan(\psi/2)$; Bondi's k-calculus
Extreme and mean ratio	Euclid II.11, VI.30	golden ratio; Fibonacci characteristic equation; Binet's formula
Sum of odd numbers	Pythagorean arithmetic, Nicomachus	square numbers as gnomons; sum of the first n cubes

Table 1: One relation, several names.

the quadratic identities are propositions of Euclid's Book II, with earlier Babylonian counterparts, and the others come from Diophantus, Brahmagupta, Fibonacci, Nicomachus, Viète, Euler and their successors, as attributed in the text. The structure that unites them is also classical. In algebra and number theory the norm identities among them are the norms of the split-complex numbers, the Gaussian integers, the Eisenstein integers, the integers of $\mathbb{Q}(\sqrt{5})$ and the quaternions and their products, the Tribonacci roots are the conjugates of a cubic field, and the remaining identities are polynomial expansions [15]. This paper adds a map. The constructions of [1–5] were developed separately, in geometry, transmission lines, rational rotations and recurrences, and present their steps without naming the identities those steps use. The paper places each identity at the step it carries in these constructions, shows the algebra of every step, and collects the steps under the classical norm structure in Section 9. Several placements are noted here in the setting of these constructions: the boundaries of the Geometric Matrix as Euclid's cut, the cubes as runs of odd intervals between triangular nodes beside the Pythagorean nodes at $4T(\ell)$, the product of consecutive node radii as four times a sum of cubes, the Fibonacci rotations as convergents in the

exact rotation path, and the quadruple $(3, 4, 12, 13)$ over the Matrix triangle. Each is an elementary consequence of a classical identity, and the paper presents them as placements, not as new results.

Purpose. The map serves four uses. First, it makes each step of the constructions auditable: a reader can check any step against a named classical identity with a recorded history, rather than against the constructions alone. Second, it identifies what the exact computations of [1] depend on. The zero norm residual of the rational rotation path follows from the two-square identity, the rotations are computed from the rotation triple with no trigonometric evaluation, and the arithmetic after a real angle is chosen is exact because the angle enters through its continued-fraction convergents; an implementation of those computations must preserve exactly these identities. A working computation does not always show which identity each of its steps depends on. When rotation matrices are composed in floating-point arithmetic, the result depends on $\cos^2\psi + \sin^2\psi = 1$, which holds there only approximately and is not named in the computation. The map names each such dependency at the step where it acts. Third, it gives the notable identities of the curriculum a reason as well as a form. They recur because they are Euclid's relations on a cut line and the norms of a small number of number systems, and the constructions show them at work across geometry, transmission lines, rotations and recurrences. Fourth, it restores the continuity of names: each identity of the school list stands beside its classical source and the names it takes later in algebra, engineering and physics (Table 1), so that the relation can be recognized under each of its names.

The order follows the objects. Section 2 lists the identities with their expansions. Section 3 treats the square-root coordinate and the difference of squares, Section 4 the cubes on triangular nodes, Section 5 the Pythagorean triples and rotations, Section 6 the Gaussian and hexagonal arithmetic, Section 7 the recurrences, and Section 8 the quadruples. Section 9 states the common structure of conjugate factors and norms, and Section 10 collects the placements.

Scope. The paper does not present new identities or new theorems. The identities are classical and are attributed in the text. The paper places each identity at a step of the cited constructions and shows the algebra of that step. The placements are algebraic: they state which identity carries which step and do not add physical claims beyond those of the cited papers. Each identity indexed by n is established symbolically for every n in its domain, through closed forms or through the recurrence and its initial values. The domains are as follows: the identities in $T(n)$ hold for $n \geq 1$ with $T(0) = 0$, and the Pythagorean nodes for $\ell \geq 1$; the Fibonacci and Lucas identities hold for $n \geq 0$, and for $n \geq 1$ wherever F_{n-1} or the ratio F_{n+1}/F_n appears; the Tribonacci power sums $K(n)$ are defined for every integer n through the recurrence run in both directions, and their identities hold for every integer n. Exact integer or rational arithmetic was used in addition to check every numerical example and the cases $n = 1$ to 39 of each indexed identity.

2 The Identities

Each identity is proved by distributing the product and collecting terms.

Square of a sum and of a difference. Euclid II.4 and II.7 [6]:

$$(a+b)^2 = (a+b)(a+b) = a^2 + ab + ba + b^2 = a^2 + 2ab + b^2, \tag{1}$$

$$(a-b)^2 = (a-b)(a-b) = a^2 - ab - ba + b^2 = a^2 - 2ab + b^2. \tag{2}$$

II.4 states (1) directly. II.7 states that the square on a whole line $a + b$ together with the square on one part a equals twice the rectangle on the whole and that part together with the square on the other part, $(a+b)^2 + a^2 = 2(a+b)a + b^2$, which is (2) with $a \to a + b$ and $b \to a$, rearranged.

Difference of squares. Euclid II.5 and II.6 [6]:

$$(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2, \qquad (a+b)(a-b) + b^2 = a^2. \tag{3}$$

II.5 cuts a line of length $2a$ at its midpoint and at a second point at distance b from it. II.6 extends a line of length $2a$ by b, and states $(2a+b)b + a^2 = (a+b)^2$, which is (3) with $a \to a+b$ and $b \to a$, since then $a+b \to 2a+b$ and $a-b \to b$.

Legendre's identities. Adding and subtracting (1) and (2), the cross terms $\pm 2ab$ cancel or double:

$$(a+b)^2 + (a-b)^2 = 2a^2 + 2b^2 = 2(a^2+b^2), \tag{4}$$

$$(a+b)^2 - (a-b)^2 = 2ab - (-2ab) = 4ab. \tag{5}$$

These are Euclid II.8 to II.10 [6]. II.9 states (4) for a line cut at its midpoint and at a second point. With $a \to a+b$ and $b \to a$, so that the sum becomes $2a+b$ and the difference becomes b, (5) reads $4(a+b)a + b^2 = (2a+b)^2$, which is II.8, and (4) reads $(2a+b)^2 + b^2 = 2\left[(a+b)^2 + a^2\right]$, which is II.10.

Product of binomials with a common term (Stevin's identity).

$$(x+a)(x+b) = x^2 + bx + ax + ab = x^2 + (a+b)x + ab. \tag{6}$$

Three-factor extension. Multiplying (6) by $(x+c)$:

$$(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc. \tag{7}$$

Cube of a sum and of a difference. Multiplying (1) by $(a+b)$:

$$(a+b)^3 = (a^2+2ab+b^2)(a+b) = a^3 + a^2b + 2a^2b + 2ab^2 + ab^2 + b^3 = a^3 + 3a^2b + 3ab^2 + b^3, \tag{8}$$

$$(a-b)^3 = (a^2-2ab+b^2)(a-b) = a^3 - a^2b - 2a^2b + 2ab^2 + ab^2 - b^3 = a^3 - 3a^2b + 3ab^2 - b^3. \tag{9}$$

Collecting $3a^2b + 3ab^2 = 3ab(a+b)$ gives Cauchy's form

$$(a+b)^3 = a^3 + b^3 + 3ab(a+b). \tag{10}$$

Sum and difference of cubes.

$$(a+b)(a^2 - ab + b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3, \tag{11}$$

$$(a-b)(a^2 + ab + b^2) = a^3 + a^2b + ab^2 - a^2b - ab^2 - b^3 = a^3 - b^3. \tag{12}$$

Fourth power of a binomial. Squaring (1):

$$\begin{aligned}(a+b)^4 &= (a^2+2ab+b^2)(a^2+2ab+b^2) \\ &= (a^4 + 2a^3b + a^2b^2) + (2a^3b + 4a^2b^2 + 2ab^3) + (a^2b^2 + 2ab^3 + b^4) \\ &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4. \end{aligned} \tag{13}$$

Collecting $4a^3b + 4ab^3 = 4ab(a^2+b^2)$ gives $(a+b)^4 = a^4 + b^4 + 4ab(a^2+b^2) + 6a^2b^2$.

Newton's binomial and the factorization of $a^n - b^n$. The coefficients of $(a+b)^n$ are the binomial coefficients $\binom{n}{k}$, which satisfy Pascal's rule $\binom{m}{k} = \binom{m-1}{k} + \binom{m-1}{k-1}$. For the difference of powers,

$$(a-b)(a^{n-1} + a^{n-2}b + \cdots + b^{n-1}) = a^n - b^n, \tag{14}$$

since every intermediate term $a^{n-k}b^k$ appears once with each sign and cancels.

Trinomial square and trinomial cube.

$$\begin{aligned}(x+y+z)^2 &= [(x+y)+z]^2 = (x+y)^2 + 2(x+y)z + z^2 \\ &= x^2 + y^2 + z^2 + 2(xy + yz + zx), \end{aligned} \tag{15}$$

$$(x+y+z)^3 - x^3 - y^3 - z^3 = 3(x+y)(y+z)(z+x). \tag{16}$$

For (16), write $e_1 = x+y+z$, $e_2 = xy+yz+zx$, $e_3 = xyz$. Each pair sum is the total minus the third variable, so by (7) with $x \to e_1$ and $(a,b,c) \to (-z,-x,-y)$,

$$(x+y)(y+z)(z+x) = (e_1 - z)(e_1 - x)(e_1 - y) = e_1^3 - e_1 \cdot e_1^2 + e_2 e_1 - e_3 = e_1 e_2 - e_3. \tag{17}$$

Expanding e_1^3 by (8) applied to $(x+y)+z$ gives $e_1^3 = x^3 + y^3 + z^3 + 3(e_1e_2 - e_3)$, which is (16).

Gauss's identity.

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx). \tag{18}$$

From the expansion of e_1^3 above, $x^3 + y^3 + z^3 - 3e_3 = e_1^3 - 3e_1e_2 = e_1(e_1^2 - 3e_2)$, and by (15), $e_1^2 - 3e_2 = x^2 + y^2 + z^2 - e_2$.

Argand's identity. By (3) with $a = x^2 + 1$ and $b = x$:

$$(x^2 + x + 1)(x^2 - x + 1) = (x^2+1)^2 - x^2 = x^4 + 2x^2 + 1 - x^2 = x^4 + x^2 + 1. \tag{19}$$

The two-square identity of Brahmagupta and Fibonacci [7, 8].

$$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2. \tag{20}$$

The right side expands to $(a^2c^2 - 2abcd + b^2d^2) + (a^2d^2 + 2abcd + b^2c^2)$, the cross terms cancel, and the four remaining terms are the expansion of the left side. Diophantus uses the composition of two sums of two squares in the Arithmetica [31], Brahmagupta states it as a rule of composition [7], and Fibonacci proves it in the Liber Quadratorum [8]. Lagrange's form $(a^2+b^2)(c^2+d^2) = (ac+bd)^2 + (ad-bc)^2$ follows with $b \to -b$.

Euler's four-square identity [9].

$$\begin{aligned}&(a_1^2 + a_2^2 + a_3^2 + a_4^2)(b_1^2 + b_2^2 + b_3^2 + b_4^2) \\ &\quad = (a_1b_1 - a_2b_2 - a_3b_3 - a_4b_4)^2 + (a_1b_2 + a_2b_1 + a_3b_4 - a_4b_3)^2 \\ &\qquad + (a_1b_3 - a_2b_4 + a_3b_1 + a_4b_2)^2 + (a_1b_4 + a_2b_3 - a_3b_2 + a_4b_1)^2. \end{aligned} \tag{21}$$

Each square on the right contributes four squared terms $a_i^2 b_j^2$, sixteen in all, which are the expansion of the left side, and the twenty-four cross terms cancel in pairs. It is the norm of the product of two quaternions [10].

The Sophie Germain identity.

$$a^4 + 4b^4 = (a^2 + 2ab + 2b^2)(a^2 - 2ab + 2b^2). \tag{22}$$

By (3) with $a \to a^2 + 2b^2$ and $b \to 2ab$: $(a^2+2b^2)^2 - (2ab)^2 = a^4 + 4a^2b^2 + 4b^4 - 4a^2b^2 = a^4 + 4b^4$.

3 The Root and the Square

3.1 The square-root coordinate

The planar map of Levi-Civita [32, 11] sets $u = \sqrt{r}\, e^{i\varphi/2}$, so that $u^2 = z = re^{i\varphi}$. Under $\varphi \to \varphi + 2\pi$ the coordinate u goes to $-u$ and u^2 returns to itself, which is the half-integer and integer rotation character of [1]. The square of the coordinate is the first step of the constructions below.

3.2 Node intervals and their accumulation

With $r_n = n^2 r_0$ and $u_n = n\sqrt{r_0}$, the difference of squares (3) gives the interval between consecutive nodes [1]:

$$\Delta r_n = u_{n+1}^2 - u_n^2 = (u_{n+1} - u_n)(u_{n+1} + u_n) = \sqrt{r_0} \cdot (2n+1)\sqrt{r_0} = (2n+1)\, r_0. \tag{23}$$

Counting from $r = 0$, the intervals are $r_0, 3r_0, 5r_0, \ldots$, and their partial sums are the nodes, $\sum_{k=1}^{n}(2k-1) = n^2$. In the hydrogen shells the same odd numbers are the dimensions $2\ell + 1$ of the angular momentum representations, which sum to the n^2 orbital states of shell n [1].

3.3 The Geometric Matrix

The Geometric Matrix [2] takes a boundary interval $[a, d]$ with centroid $b = (a + d)/2$ and span $c = d - a$, so that $a = b - c/2$ and $d = b + c/2$. The two boundaries are the two extents of Euclid's cut, with half-length b and offset $c/2$, and (3) gives

$$a\, d = \left(b - \tfrac{c}{2}\right)\left(b + \tfrac{c}{2}\right) = b^2 - \left(\tfrac{c}{2}\right)^2, \qquad a\, d + \left(\tfrac{c}{2}\right)^2 = b^2. \tag{24}$$

For the rope boundary $[1, 5]$: $b = 3$, $c = 4$, and $1 \cdot 5 + 2^2 = 5 + 4 = 9 = 3^2$. The equilibrium condition $\Delta = b^2 + c^2 - d^2 = 0$ of [2] selects $d = 5a$, and the boundary $[n, 5n]$ gives the triangle $(b, c, d) = (3n, 4n, 5n)$.

3.4 The Smith chart

With incident amplitude a and reflected amplitude b on a lossless line, the extents $a + b$ and $a - b$ of II.5 are V and $Z_0 I$, their ratio $u = (a + b)/(a - b)$ is the normalized impedance, and the map $\Gamma = (u - 1)/(u + 1)$ returns the reflection coefficient b/a [3]. Proposition II.5 multiplied by $1/(2Z_0)$ is the power balance $P_{\text{net}} + P_{\text{ref}} = P_{\text{inc}}$ [3], and for complex phasor amplitudes the net power is proportional to $\text{Re}\left[(a + b)\overline{(a - b)}\right] = |a|^2 - |b|^2$, since the cross terms $-a\bar{b} + \bar{a}b$ are purely imaginary [3]. For complex normalized impedance $z = r + jx$, the modulus of Γ satisfies

$$1 - |\Gamma|^2 = \frac{(r+1)^2 - (r-1)^2}{(r+1)^2 + x^2} = \frac{4r}{(r+1)^2 + x^2}, \tag{25}$$

where the numerator is Legendre's identity (5) with $a = r$ and $b = 1$: $(r^2 + 2r + 1) - (r^2 - 2r + 1) = 4r$. The disk $|\Gamma| < 1$ is therefore exactly the half-plane $r > 0$.

3.5 Composition of two boundary stages

In the frequency composition of [5], two boundary stages, each projected to the first-order factor $1 + \lambda\Delta t$, compose to

$$f_0(1 + \lambda\Delta t)^2 = f_0\left(1 + 2\lambda\Delta t + \lambda^2\Delta t^2\right), \tag{26}$$

which is the square of a sum (1) with $a = 1$ and $b = \lambda\Delta t$. The quadratic term $\lambda^2\Delta t^2$ is the term b^2 of the identity.

4 Cubes on Triangular Nodes

Let $T(n) = n(n+1)/2$ be the n-th triangular number, with $T(0) = 0$. The run of odd intervals from node $T(n-1)$ to node $T(n)$ has n members and sums to a cube:

n	intervals in the run	from node to node	sum
1	1	$0 \to 1$	$1 = 1^3$
2	3, 5	$1 \to 3$	$8 = 2^3$
3	7, 9, 11	$3 \to 6$	$27 = 3^3$
4	13, 15, 17, 19	$6 \to 10$	$64 = 4^3$

This is the arrangement of Nicomachus [12]. Since accumulated intervals equal the node radius, the run sums to $\left[T(n)^2 - T(n-1)^2\right] r_0$, and (3) gives

$$T(n) - T(n-1) = \frac{n(n+1)}{2} - \frac{(n-1)n}{2} = \frac{n\left[(n+1) - (n-1)\right]}{2} = n, \tag{27}$$

$$T(n) + T(n-1) = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} = \frac{n\left[(n+1) + (n-1)\right]}{2} = n^2, \tag{28}$$

$$T(n)^2 - T(n-1)^2 = n \cdot n^2 = n^3. \tag{29}$$

The width of the run is n, its center is n^2, and the cube is their product. Summing over n, consecutive terms cancel:

$$1^3 + 2^3 + \cdots + n^3 = \left[T(1)^2 - T(0)^2\right] + \cdots + \left[T(n)^2 - T(n-1)^2\right] = T(n)^2 = \left(\frac{n(n+1)}{2}\right)^2. \tag{30}$$

The sum of the first n cubes is the square on the triangular number $T(n)$, for example $1+8+27+64 = 100 = 10^2$.

The sum of cubes appears in the adjacent-node wavelength of [1], $\lambda_n = n^2(n+1)^2 r_0/(2n+1)$. Since $n^2(n+1)^2 = (2T(n))^2 = 4T(n)^2$,

$$r_n r_{n+1} = n^2(n+1)^2 r_0^2 = 4\left(1^3 + 2^3 + \cdots + n^3\right) r_0^2, \qquad \lambda_n = \frac{4\left(1^3 + \cdots + n^3\right)}{2n+1}\, r_0. \tag{31}$$

At $n = 1$: $4 \cdot 1/3 = 4/3 = 1^2 \cdot 2^2/3$. At $n = 2$: $4 \cdot 9/5 = 36/5 = 2^2 \cdot 3^2/5$.

5 Triples and Rotations

5.1 The rotation triple

Trigonometric expressions in this paper state the conventional reading of a rational quantity, so that a reader working in trigonometric coordinates can locate the same step; no construction step evaluates a trigonometric function. For a rational parameter $t = p/q$, the rotation entries of [1] are

$$e^{i\psi} = \frac{(1+it)^2}{1+t^2} = \frac{(q+ip)^2}{q^2+p^2} = \frac{(q^2-p^2)+i\,2pq}{q^2+p^2}, \qquad t = \tan\frac{\psi}{2}. \tag{32}$$

Legendre's identity (5) with $a = q^2$ and $b = p^2$ gives the norm:

$$(q^2+p^2)^2 - (q^2-p^2)^2 = 4p^2q^2 = (2pq)^2, \qquad (q^2-p^2)^2 + (2pq)^2 = (q^2+p^2)^2. \tag{33}$$

The parameter $t = 1/2$ gives $(3, 4, 5)$, which is the triangle of the Geometric Matrix boundary [1, 5] [2].

5.2 Pythagorean nodes

When the odd interval $2n + 1$ is itself a square m^2, the difference of squares (3) on consecutive nodes reads $m^2 + n^2 = (n + 1)^2$. With $m = 2\ell + 1$,

$$n = \frac{m^2-1}{2} = \frac{4\ell^2+4\ell+1-1}{2} = 2\ell(\ell+1) = 4\,T(\ell), \tag{34}$$

which gives the nodes $4, 12, 24, 40, \ldots$ and the triples $(m, n, n + 1)$, which are $(3, 4, 5)$, $(5, 12, 13)$, $(7, 24, 25)$, $(9, 40, 41)$ and so on, derived in [1]. These nodes are referred to here as Pythagorean nodes. The relationship is the classical rule, attributed to Pythagoras by Proclus [13], that sends an odd m to the triple $(m, (m^2 - 1)/2, (m^2 + 1)/2)$, and it gives the family of triples whose hypotenuse exceeds the longer leg by 1. The Pythagorean nodes lie at four times the triangular numbers, and the run boundaries of Section 4 lie at the triangular numbers themselves.

5.3 Composition of rotations

Two rotations $c_1 + is_1$ and $c_2 + is_2$ compose by the product rule $(c_1c_2 - s_1s_2) + i(c_1s_2 + s_1c_2)$ of [1], and the two-square identity (20) states that the norm of the product is the product of the norms. The rotation reading is that of complex multiplication; the identity itself is the classical algebraic statement:

$$(c_1^2 + s_1^2)(c_2^2 + s_2^2) = (c_1c_2 - s_1s_2)^2 + (c_1s_2 + s_1c_2)^2. \tag{35}$$

In the parameter t, the product of $1 + it_1$ and $1 + it_2$ is $(1 - t_1t_2) + i(t_1 + t_2)$, and the ratio of imaginary to real part gives the tangent addition rule $t_3 = (t_1 + t_2)/(1 - t_1t_2)$ of [1]. The norm residual of the exact rotation path of [1] is zero because of this identity.

5.4 The Hadamard step

The Hadamard step of [1] sends the amplitude pair (a, b) to the extents $(a + b, a - b)$. For complex amplitudes Legendre's identity (4) reads $|a + b|^2 + |a - b|^2 = 2(|a|^2 + |b|^2)$, so the update doubles

the sum of squares, which is the common factor $\sqrt{2}$ that [1] drops. The same identity gives the denominator of the X-basis probability of [1],

$$P(+) = \frac{|a+b|^2}{2\,(|a|^2+|b|^2)}. \tag{36}$$

For $a = (3-4i)/5$ and $b = (3+4i)/5$: $|a+b|^2 + |a-b|^2 = 36/25 + 64/25 = 100/25 = 4 = 2(1+1)$.

5.5 Rational Doppler factors

The relativistic Doppler factor is $k = \sqrt{(1+\beta)/(1-\beta)}$. Solving for β:

$$k^2(1-\beta) = 1+\beta \implies k^2 - 1 = \beta(k^2+1) \implies \beta = \frac{k^2-1}{k^2+1}, \tag{37}$$

which is the Smith map $\Gamma = (u-1)/(u+1)$ of [3] with $u = k^2$. For $k = q/p$, $\beta = (q^2-p^2)/(q^2+p^2)$, and by (3)

$$1-\beta^2 = \frac{(q^2+p^2)^2 - (q^2-p^2)^2}{(q^2+p^2)^2} = \frac{(2p^2)(2q^2)}{(q^2+p^2)^2}, \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}} = \frac{q^2+p^2}{2pq}. \tag{38}$$

For $k = 2$: $\beta = 3/5$ and $\gamma = 5/4$. For $k = 3/2$: $\beta = 5/13$ and $\gamma = 13/12$. Rational Doppler factors carry the rotation triple; this is Bondi's k-calculus [14].

6 Gaussian and Hexagonal Arithmetic

6.1 The Sophie Germain identity at $e^{i\pi/4}$

Over the Gaussian integers the Sophie Germain identity (22) is the difference of squares applied twice. First,

$$(1+i)^2 = 1 + 2i + i^2 = 2i, \qquad (1+i)^4 = (2i)^2 = -4, \tag{39}$$

so $4b^4 = -(1+i)^4 b^4$ and $a^4 + 4b^4 = a^4 - ((1+i)b)^4$. With $w = (1+i)b$ and $w^2 = 2ib^2$, (3) gives

$$a^4 - w^4 = (a^2 - w^2)(a^2 + w^2) = (a^2 - 2ib^2)(a^2 + 2ib^2). \tag{40}$$

Applying (3) again and pairing conjugate factors,

$$(a - (1+i)b)\,(a - (1-i)b) = a^2 - 2ab + 2b^2, \tag{41}$$

$$(a + (1+i)b)\,(a + (1-i)b) = a^2 + 2ab + 2b^2, \tag{42}$$

whose product is (22). The unit $(1+i)/\sqrt{2} = e^{i\pi/4}$ has fourth power $-4/(\sqrt{2})^4 = -1$ and eighth power 1. This is the value $e^{i\pi/4}$ computed exactly in [1], whose fourth power is -1 and eighth power is 1, and its fourth power $e^{i\pi} = -1$ is the sign of the half-angle factor at a rotation of 2π in the double cover of [1], with the identity restored at 4π. The length of $1+i$ is the factor $\sqrt{2}$ of the Hadamard step.

6.2 The inscribed hexagon

Divide a circle of radius r into six equal arcs. Each central angle is $360^\circ/6 = 60^\circ$, and the triangle formed by the center and two adjacent vertices has two sides r enclosing 60°, so its base angles are $(180^\circ - 60^\circ)/2 = 60^\circ$. The triangle is equilateral and each side of the hexagon equals r, which is Euclid IV.15 [6]. The perimeter is

$$6r = 3 \cdot 2r = 3D. \tag{43}$$

On the unit circle the six vertices are the roots of $x^6 - 1 = 0$. The difference of cubes (12) with $a = x^2$ and $b = 1$, followed by the difference of squares and Argand's identity (19), splits them:

$$x^6 - 1 = (x^2)^3 - 1^3 = (x^2 - 1)(x^4 + x^2 + 1), \tag{44}$$

$$x^2 - 1 = (x - 1)(x + 1), \qquad x^4 + x^2 + 1 = (x^2 + x + 1)(x^2 - x + 1). \tag{45}$$

Factor	Roots	Vertices
$x - 1$	1	0°
$x + 1$	-1	180°
$x^2 + x + 1$	$e^{\pm 2\pi i/3}$	$120^\circ,\ 240^\circ$
$x^2 - x + 1$	$e^{\pm i\pi/3}$	$60^\circ,\ 300^\circ$

The vertices at $\pm 60^\circ$ and $\pm 120^\circ$ have coordinates with $\sqrt{3}/2$, and their exact arithmetic is that of the Eisenstein integers, whose norm $a^2 - ab + b^2$ is the second factor of the sum of cubes (11).

6.3 From the hexagon to 22/7

The ratio of the hexagon perimeter to the diameter is 3. The continued fraction of π begins $[3; 7, 15, 1, 292]$, with convergents

$$\frac{3}{1}, \quad 3 + \frac{1}{7} = \frac{22}{7}, \quad \frac{333}{106}, \quad \frac{355}{113}. \tag{46}$$

The hexagon ratio 3 is the first convergent and 22/7 is the second. The error of 22/7 is $22/7 - \pi = 0.00126$, below the convergent bound $1/(q_k q_{k+1}) = 1/(7 \cdot 106) = 0.00135$ [15]. Archimedes began from the inscribed and circumscribed hexagons and doubled the number of sides to 96, proving $223/71 < \pi < 22/7$ [16]. A real angle enters the exact rotation path of [1] through the convergents of its parameter t with error at most $2/q^2$, and 22/7 is the same kind of object for the ratio of circumference to diameter.

7 Recurrences

7.1 The two roots of the Fibonacci equation

The Fibonacci recurrence $F_{n+1} = F_n + F_{n-1}$ has the characteristic equation $x^2 - x - 1 = 0$. Writing it as $(x - \varphi)(x - \psi)$ and comparing with (6) gives

$$\varphi + \psi = 1, \qquad \varphi\psi = -1, \qquad \varphi - \psi = \sqrt{5}, \tag{47}$$

with $\varphi = (1+\sqrt{5})/2$, $\psi = (1-\sqrt{5})/2$, and $(\varphi-\psi)^2 = (\varphi+\psi)^2 - 4\varphi\psi = 1+4 = 5$ by (5). The product $\varphi\psi = -1$ is the source of every alternating sign below.

Euclid II.11 and VI.30. Euclid II.11 cuts a line of length a at a point so that the rectangle on the whole line and one part equals the square on the other part [6]. With the other part of length x, the condition is $a(a-x) = x^2$, and with $a = 1$,

$$x^2 + x - 1 = 0, \qquad x = \frac{-1+\sqrt{5}}{2} = \frac{1}{\varphi}. \tag{48}$$

Dividing $x^2 + x - 1 = 0$ by x^2 gives $1 + 1/x - 1/x^2 = 0$, and with $y = 1/x$ this is $y^2 - y - 1 = 0$, the Fibonacci equation. Euclid VI.30 states the same cut as division in extreme and mean ratio [6]. The ratio φ of the Fibonacci recurrence is the reciprocal of Euclid's cut.

Binet's formula as a difference of powers. By (14) with $a = \varphi$ and $b = \psi$,

$$F_n = \frac{\varphi^n - \psi^n}{\varphi - \psi} = \varphi^{n-1} + \varphi^{n-2}\psi + \cdots + \psi^{n-1}, \tag{49}$$

which is Binet's formula [17]. The closed form satisfies the recurrence, since multiplying $\varphi^2 = \varphi + 1$ by φ^{n-1} gives $\varphi^{n+1} = \varphi^n + \varphi^{n-1}$, and likewise for ψ. It takes the values 0 at $n = 0$ and $(\varphi-\psi)/\sqrt{5} = 1$ at $n = 1$. A sequence is fixed by the recurrence and two initial values, so the closed form equals F_n for every $n \geq 0$. Define the Lucas numbers $L_n = \varphi^n + \psi^n$ [18]; by the same argument they satisfy $L_{n+1} = L_n + L_{n-1}$ with $L_0 = 2$ and $L_1 = \varphi + \psi = 1$. Setting $x = \varphi^n$ and $y = \psi^n$ gives $xy = (\varphi\psi)^n = (-1)^n$, $x - y = \sqrt{5}F_n$ and $x + y = L_n$.

Lucas and Cassini. Legendre's identity (5) gives

$$L_n^2 - 5F_n^2 = (x+y)^2 - (x-y)^2 = 4xy = 4(-1)^n. \tag{50}$$

For Cassini's identity [19, 20], expand with Binet:

$$\begin{aligned} 5\left(F_{n+1}F_{n-1} - F_n^2\right) &= (\varphi^{n+1} - \psi^{n+1})(\varphi^{n-1} - \psi^{n-1}) - (\varphi^n - \psi^n)^2 \\ &= -(\varphi\psi)^{n-1}(\varphi^2 + \psi^2) + 2(\varphi\psi)^n. \end{aligned} \tag{51}$$

By (1), $\varphi^2 + \psi^2 = (\varphi+\psi)^2 - 2\varphi\psi = 1 + 2 = 3$, and $(\varphi\psi)^n = -(\varphi\psi)^{n-1}$, so the right side is $(-1)^{n-1}(-3-2) = 5(-1)^n$, and

$$F_{n+1}F_{n-1} - F_n^2 = (-1)^n. \tag{52}$$

The ratio of consecutive terms. For $n \geq 1$, the ratio $r(n) = F_{n+1}/F_n$ approaches φ with alternating sign [4]. From Binet,

$$F_{n+1} - \varphi F_n = \frac{\varphi^{n+1} - \psi^{n+1} - \varphi^{n+1} + \varphi\psi^n}{\sqrt{5}} = \frac{\psi^n(\varphi - \psi)}{\sqrt{5}} = \psi^n, \tag{53}$$

so

$$r(n) - \varphi = \frac{\psi^n}{F_n} = \frac{(-1)^n}{\varphi^n F_n}, \qquad r(n) - \varphi \approx \frac{(-1)^n\sqrt{5}}{\varphi^{2n}}, \tag{54}$$

using $F_n \approx \varphi^n/\sqrt{5}$ for large n. At $n = 10$ the difference is 1.478×10^{-4}, and the approximation gives 1.478×10^{-4}.

7.2 Fibonacci rotations

Sum of two consecutive squares. With $1+\varphi^2=\varphi+2=\sqrt{5}\varphi$ and $1+\psi^2=\psi+2=-\sqrt{5}\psi$ (from $\varphi^2=\varphi+1$, $\psi^2=\psi+1$, $\sqrt{5}=2\varphi-1=1-2\psi$),

$$\begin{aligned} 5\left(F_n^2+F_{n+1}^2\right) &= \varphi^{2n}(1+\varphi^2)+\psi^{2n}(1+\psi^2)-2(\varphi\psi)^n(1+\varphi\psi) \\ &= \sqrt{5}\,\varphi^{2n+1}-\sqrt{5}\,\psi^{2n+1}-0, \end{aligned} \tag{55}$$

and dividing by $5=\sqrt{5}\cdot\sqrt{5}$ gives $F_n^2+F_{n+1}^2=(\varphi^{2n+1}-\psi^{2n+1})/\sqrt{5}=F_{2n+1}$, one of the identities collected by Lucas [18].

The rotation triple. Set $t=F_n/F_{n+1}$ in the rotation of Section 5.1, with $p=F_n$ and $q=F_{n+1}$. By (3),

$$q^2-p^2=(F_{n+1}-F_n)(F_{n+1}+F_n)=F_{n-1}F_{n+2}, \qquad 2pq=2F_nF_{n+1}, \qquad q^2+p^2=F_{2n+1}. \tag{56}$$

n	$t=F_n/F_{n+1}$	$(F_{n-1}F_{n+2},\ 2F_nF_{n+1},\ F_{2n+1})$
1	1/1	$(0,2,2)$
2	1/2	$(3,4,5)$
3	2/3	$(5,12,13)$
4	3/5	$(16,30,34)$
5	5/8	$(39,80,89)$

The identity $F_{n+1}^2-F_n^2=F_{n-1}F_{n+2}$ and the Pythagorean triangles built from consecutive Fibonacci numbers are classical [33]; the placement here is their role in the exact rotation path. The ratios F_n/F_{n+1} are the continued-fraction convergents of $1/\varphi=[0;1,1,1,\dots]$, so the Fibonacci rotations are the convergent sequence of the angle $\psi_\infty=2\arctan(1/\varphi)$, each within $2/F_{n+1}^2$ by the bound of [1]. The limit has the exact cosine

$$\cos\psi_\infty=\frac{1-1/\varphi^2}{1+1/\varphi^2}=\frac{\varphi^2-1}{\varphi^2+1}=\frac{\varphi}{\varphi+2}=\frac{\varphi}{\sqrt{5}\,\varphi}=\frac{1}{\sqrt{5}}. \tag{57}$$

7.3 Powers of the two roots

With $x=\varphi^n$, $y=\psi^n$ and $xy=(-1)^n$:

Square of a sum. $x^2+y^2=(x+y)^2-2xy$ gives $L_{2n}=L_n^2-2(-1)^n$. At $n=2$: $L_4=7=3^2-2$.

Difference of squares. $x^2-y^2=(x-y)(x+y)$ gives $F_{2n}=F_nL_n$. At $n=4$: $F_8=21=3\cdot 7$.

Cauchy's cube. By (10), $x^3+y^3=(x+y)^3-3xy(x+y)$, so $L_{3n}=L_n^3-3(-1)^nL_n$. At $n=2$: $L_6=18=27-9$.

Fourth power. By (13), $(x+y)^4=x^4+y^4+4xy(x^2+y^2)+6x^2y^2$, so $L_n^4=L_{4n}+4(-1)^nL_{2n}+6$. At $n=2$: $81=47+28+6$.

Sum and difference of cubes. By (12),

$$F_{3n}=\frac{x^3-y^3}{\sqrt{5}}=\frac{(x-y)(x^2+xy+y^2)}{\sqrt{5}}=F_n\left(L_{2n}+(-1)^n\right)=F_n\left(5F_n^2+3(-1)^n\right), \tag{58}$$

using $L_{2n} = L_n^2 - 2(-1)^n = 5F_n^2 + 4(-1)^n - 2(-1)^n$. The cube identity

$$F_{3n} = F_{n+1}^3 + F_n^3 - F_{n-1}^3 \tag{59}$$

follows. By (12) with $F_{n+1} - F_{n-1} = F_n$,

$$F_{n+1}^3 - F_{n-1}^3 + F_n^3 = F_n \left[F_{n+1}^2 + F_{n+1}F_{n-1} + F_{n-1}^2 + F_n^2 \right]. \tag{60}$$

By Cassini, $F_{n+1}F_{n-1} = F_n^2 + (-1)^n$, and by (2), $F_{n+1}^2 + F_{n-1}^2 = (F_{n+1} - F_{n-1})^2 + 2F_{n+1}F_{n-1} = 3F_n^2 + 2(-1)^n$. The bracket is $3F_n^2 + 2(-1)^n + F_n^2 + (-1)^n + F_n^2 = 5F_n^2 + 3(-1)^n$, which matches. At $n = 4$: $F_{12} = 144 = 125 + 27 - 8$. Squares of consecutive terms land on the index $2n + 1$, and cubes on the index $3n$.

Pascal's triangle. Let $G(n) = \sum_k \binom{n-1-k}{k}$, the sum along a shallow diagonal. By Pascal's rule,

$$G(n+1) = \sum_k \binom{n-k}{k} = \sum_k \left[\binom{n-1-k}{k} + \binom{n-1-k}{k-1} \right] = G(n) + G(n-1), \tag{61}$$

with $G(1) = G(2) = 1$, so $G(n) = F_n$. At $n = 6$: $\binom{5}{0} + \binom{4}{1} + \binom{3}{2} = 1 + 4 + 3 = 8$.

7.4 The three roots of the Tribonacci equation

The Tribonacci equation $x^3 - x^2 - x - 1 = 0$ has roots α, β, γ. Writing it as $(x - \alpha)(x - \beta)(x - \gamma)$ and comparing with the three-factor extension (7) of Stevin's identity,

$$\alpha + \beta + \gamma = 1, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = -1, \qquad \alpha\beta\gamma = 1. \tag{62}$$

Define $K(n) = \alpha^n + \beta^n + \gamma^n$. Each root satisfies $x^3 = x^2 + x + 1$, so multiplying by x^{n-3} gives $x^n = x^{n-1} + x^{n-2} + x^{n-3}$ for every integer n, the roots being nonzero since $\alpha\beta\gamma = 1$. Summing over the three roots, $K(n) = K(n-1) + K(n-2) + K(n-3)$, with $K(0) = 3$, $K(1) = 1$ and $K(2) = (\alpha+\beta+\gamma)^2 - 2(\alpha\beta+\beta\gamma+\gamma\alpha) = 1+2 = 3$ by (15), giving $3, 1, 3, 7, 11, 21, 39, 71, 131, 241, \ldots$. Running the recurrence backward gives $K(-1) = -1$, $K(-2) = -1$, $K(-3) = 5$, the power sums of the reciprocal roots.

Set $x = \alpha^n$, $y = \beta^n$, $z = \gamma^n$. Then $x + y + z = K(n)$, $xyz = (\alpha\beta\gamma)^n = 1$, and

$$xy + yz + zx = (\alpha\beta\gamma)^n \left(\gamma^{-n} + \alpha^{-n} + \beta^{-n} \right) = K(-n). \tag{63}$$

Gauss's identity. By (18) and (15), $x^2 + y^2 + z^2 - (xy + yz + zx) = K(n)^2 - 3K(-n)$, so

$$K(3n) - 3 = K(n) \left(K(n)^2 - 3K(-n) \right). \tag{64}$$

At $n = 1$: $K(3) = 7 = 3 + 1 \cdot (1 + 3)$. At $n = 2$: $K(6) = 39 = 3 + 3 \cdot (9 + 3)$.

Trinomial cube. By (16) and (17),

$$K(n)^3 - K(3n) = 3 \left(K(n)K(-n) - 1 \right). \tag{65}$$

At $n = 2$: $27 - 39 = -12 = 3\,(3 \cdot (-1) - 1)$. Gauss's identity and the trinomial cube give the same relation, the first through the sum of cubes and the second through the cube of the sum. It is the three-root form of $L_{3n} = L_n^3 - 3(-1)^n L_n$.

8 Quadruples

8.1 The four-square step

The two-square identity (20) composes rotations in the plane, and Euler's four-square identity (21) is the norm of quaternion multiplication [9, 10], which composes rotations in space. For integers a, b, c, d, set $P = a^2 + b^2$, $Q = c^2 + d^2$ and

$$x = P - Q, \qquad y = 2(ad + bc), \qquad z = 2(bd - ac), \qquad w = P + Q. \tag{66}$$

Legendre's identity (5) gives $w^2 - x^2 = 4PQ$. The two-square identity gives

$$(ad+bc)^2 + (bd-ac)^2 = a^2d^2 + 2abcd + b^2c^2 + b^2d^2 - 2abcd + a^2c^2 = (a^2+b^2)(c^2+d^2) = PQ, \tag{67}$$

so $y^2 + z^2 = 4PQ = w^2 - x^2$ and

$$x^2 + y^2 + z^2 = w^2, \tag{68}$$

a Pythagorean quadruple. This parametrization of the quadruples is known as Lebesgue's identity [34]. The quaternion $a + bi + cj + dk$ defines a rotation of space through the matrix of Euler and Rodrigues [21], whose entries are quadratic forms in a, b, c, d divided by w, and $(x, y, z)/w$ is the first column of that matrix, the image of the unit vector $(1, 0, 0)$ under the rotation. That column is $(a^2 + b^2 - c^2 - d^2,\ 2(bc + ad),\ 2(bd - ac))\ /w$, which is $(x, y, z)/w$ term by term. The rotation and the rational point on the unit sphere are exact, with no trigonometric function. Setting $b = c = 0$ gives $(a^2 - d^2,\ 2ad,\ 0,\ a^2 + d^2)$, the planar triple of Section 5.1 with $(q, p) = (a, d)$. The map of Kustaanheimo and Stiefel [22] lifts the planar square-root map of Section 3.1 to three dimensions through this quaternion arithmetic.

(a, b, c, d)	(x, y, z, w)
$(2, 0, 0, 1)$	$(3, 4, 0, 5)$
$(1, 1, 0, 1)$	$(1, 2, 2, 3)$
$(2, 1, 1, 0)$	$(4, 2, -4, 6)$
$(2, 2, 1, 2)$	$(3, 12, 4, 13)$

8.2 Closing the space diagonal over the 3-4-5 triangle

Take the equilibrium triangle of the Geometric Matrix, legs $b = 3n$ and $c = 4n$ with hypotenuse $d = 5n$ [2], and extend it orthogonally to a depth h. The space diagonal s of the box satisfies $s^2 = b^2 + c^2 + h^2 = d^2 + h^2$, so the face diagonal d becomes a leg of a second right triangle. For s to be an integer, (3) requires

$$(s - h)(s + h) = d^2 = 25n^2. \tag{69}$$

At $n = 1$ the positive factor pairs of 25 are $(1, 25)$ and $(5, 5)$. The pair $(1, 25)$ gives $2h = 24$, so $h = 12$ and $s = 13$; the pair $(5, 5)$ gives $h = 0$. The depth 12 is the only positive integer depth that closes the diagonal over the 3-4-5 triangle:

$$3^2 + 4^2 + 12^2 = 9 + 16 + 144 = 169 = 13^2. \tag{70}$$

For scale n, the depth $12n$ closes it with diagonal $13n$, since $9n^2 + 16n^2 + 144n^2 = 169n^2$. The quadruple $(3, 4, 12, 13)$ is the triple $(3, 4, 5)$ followed by the triple $(5, 12, 13)$ on its hypotenuse, and the four-square step generates it from $(a, b, c, d) = (2, 2, 1, 2)$, as in the table above.

9 Conjugate Factors and Norms

Two kinds of identity carry the steps of Sections 3 to 8. The first kind are norm relations: a quantity is written as the product of an element and its conjugate, which is the difference of squares (3) taken in a number system where a conjugate is defined, and the product $N(z) = z\,\bar{z}$ is the norm of z [15]. The difference of squares, the two-square and four-square identities, the cube factorizations and the Fibonacci relation $L_n^2 - 5F_n^2 = 4(-1)^n$ are of this kind. The second kind are polynomial expansions: the binomial powers, Cauchy's cube, Stevin's identity and its three-factor extension, Newton's binomial, the trinomial square and cube, and Gauss's identity. They carry the relations among powers and symmetric functions of conjugate roots in Section 7. The two kinds meet in the recurrences, where the expansions act on conjugates.

The split-complex numbers. Ordinary integer arithmetic has no nontrivial conjugation, and over the integers (3) is a factorization. It becomes a norm in the split-complex numbers $a+bj$ with $j^2 = 1$ and conjugate $a - bj$ [35]:

$$N(a+bj) = (a+bj)(a-bj) = a^2 - b^2 j^2 = a^2 - b^2. \tag{71}$$

The multiplication commutes, so N is multiplicative. With $(a+bj)(c+dj) = (ac+bd)+(ad+bc)j$, this gives $(a^2-b^2)(c^2-d^2) = (ac+bd)^2 - (ad+bc)^2$, which is Brahmagupta's composition for forms $x^2 - Ny^2$ [7], taken here at $N = 1$; Section 9 meets the same form again at $N = 5$. As a factorization over the integers, (3) carries the node intervals, the Matrix boundaries, the Smith chart ratio, the cubes on triangular nodes and the closure of the space diagonal.

The Gaussian integers. The conjugate of $a + ib$ is $a - ib$, and by (3)

$$N(a+ib) = (a+ib)(a-ib) = a^2 - (ib)^2 = a^2 + b^2. \tag{72}$$

Since multiplication commutes, $N(zw) = zw\,\bar{z}\bar{w} = (z\bar{z})(w\bar{w}) = N(z)N(w)$. With $z = a + ib$ and $w = c + id$, the product is $zw = (ac - bd) + i(ad + bc)$, and $N(zw) = N(z)N(w)$ is the two-square identity (20), which composes the rotations of Section 5.3. The Sophie Germain identity of Section 6.1 is the same splitting taken twice at the element $1 + i$.

The Eisenstein integers. Let $\omega = e^{2\pi i/3}$, so that $\omega^2 + \omega + 1 = 0$, $\omega + \bar{\omega} = -1$ and $\omega\bar{\omega} = 1$. Then

$$N(a+b\omega) = (a+b\omega)(a+b\bar{\omega}) = a^2 + ab(\omega+\bar{\omega}) + b^2\omega\bar{\omega} = a^2 - ab + b^2, \tag{73}$$

and $(a+b)(a+b\omega)(a+b\bar{\omega}) = (a+b)(a^2-ab+b^2) = a^3+b^3$, the sum of cubes (11); with $b \to -b$, $(a-b\omega)(a-b\bar{\omega}) = a^2+ab+b^2$, and $(a-b)(a-b\omega)(a-b\bar{\omega}) = (a-b)(a^2+ab+b^2) = a^3-b^3$, the difference of cubes (12). These are the factors that split the vertices of the inscribed hexagon in Section 6.2.

The integers of $\mathbb{Q}(\sqrt{5})$. The ring of integers of $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[\varphi]$ with $\varphi = (1+\sqrt{5})/2$. The conjugate of $\sqrt{5}$ is $-\sqrt{5}$, so the conjugate of φ is ψ and $N(\varphi) = \varphi\psi = -1$. Since $\varphi^n = (L_n + \sqrt{5}F_n)/2$,

$$N(\varphi^n) = \frac{L_n + \sqrt{5}F_n}{2} \cdot \frac{L_n - \sqrt{5}F_n}{2} = \frac{L_n^2 - 5F_n^2}{4} = N(\varphi)^n = (-1)^n, \tag{74}$$

which is the relation $L_n^2 - 5F_n^2 = 4(-1)^n$ of Section 7.1, and every alternating sign of Section 7 is a power of the norm $N(\varphi) = -1$. The norm of $\mathbb{Q}(\sqrt{5})$ is Brahmagupta's form $x^2 - 5y^2$ [7], applied to $L_n/2$ and $F_n/2$.

The cubic field of the Tribonacci roots. The roots α, β, γ of $x^3 - x^2 - x - 1$ are the three algebraic conjugates of α, the images of α under the three embeddings of the cubic field $\mathbb{Q}(\alpha)$

into the complex numbers, and the field norm of α is their product $N(\alpha) = \alpha\beta\gamma = 1$ [15]. The relations of Section 7.4 are relations among the power sums of these conjugates, and the polynomial expansions of the second kind carry them.

The quaternions. The conjugate of $q = a + bi + cj + dk$ is $\bar{q} = a - bi - cj - dk$, and $N(q) = q\bar{q} = a^2 + b^2 + c^2 + d^2$ [10]. Multiplication does not commute, since $ij = k = -ji$, and the conjugate of a product reverses the order of the factors, $\overline{pq} = \bar{q}\,\bar{p}$. Since $N(q)$ is real it commutes with every quaternion, so

$$N(pq) = pq\,\bar{q}\,\bar{p} = p\,N(q)\,\bar{p} = N(p)N(q). \tag{75}$$

Writing $p = a_1 + a_2 i + a_3 j + a_4 k$ and $q = b_1 + b_2 i + b_3 j + b_4 k$, the four components of pq are the four bracketed terms of (21), and $N(pq) = N(p)N(q)$ is Euler's four-square identity, which carries the quadruples of Section 8. The quaternions are the one system in this paper where the order of the factors matters.

The identities of the catalogue appear in the constructions because the constructions split quantities into conjugate parts, multiply such splittings, and expand powers and products of conjugates. The norm structure and the polynomial identities are classical [15]. What Sections 3 to 8 add is the location of each use: which identity, in which number system, at which step of which construction.

10 Placements

Each row of the table is a check. A reader can take the step in the third column, turn to the section in the fourth, and verify the step against the identity in the first column and, where one applies, the proposition of Euclid in the second.

Identity	Euclid	Step it carries	Section
Square of a sum and difference	II.4, II.7	composition of two boundary stages; $L_{2n} = L_n^2 - 2(-1)^n$; $\varphi^2 + \psi^2 = 3$	3.5, 7
Difference of squares	II.5, II.6	node intervals; Matrix boundaries $ad + (c/2)^2 = b^2$; Smith ratio and power balance; $n^3 = T(n)^2 - T(n-1)^2$; Pythagorean nodes; $F_{2n} = F_n L_n$; rational γ; closure $(s-h)(s+h) = d^2$	3, 4, 5, 7, 8
Legendre	II.8, II.9, II.10	rotation triple; Smith $1 - \lvert\Gamma\rvert^2$; Hadamard $\sqrt{2}$ and $P(+)$; $L_n^2 - 5F_n^2 = 4(-1)^n$; $w^2 - x^2 = 4PQ$	3.4, 5, 7, 8
Common term (Stevin)		$\varphi + \psi = 1$, $\varphi\psi = -1$	7.1
Three-factor extension		$\alpha + \beta + \gamma = 1$, $\alpha\beta + \beta\gamma + \gamma\alpha = -1$, $\alpha\beta\gamma = 1$	7.4
Cube of a sum (Cauchy)		$L_{3n} = L_n^3 - 3(-1)^n L_n$	7.3
Sum and difference of cubes		$F_{3n} = F_{n+1}^3 + F_n^3 - F_{n-1}^3$; $x^6 - 1$; Eisenstein norm	6.2, 7.3
Fourth power		$L_n^4 = L_{4n} + 4(-1)^n L_{2n} + 6$	7.3
Newton and $a^n - b^n$		Binet's formula; Pascal diagonals	7.1, 7.3
Trinomial square		$K(2) = 3$; Gauss form for $K(3n)$	7.4
Trinomial cube		$K(n)^3 - K(3n) = 3(K(n)K(-n) - 1)$	7.4
Gauss		$K(3n) - 3 = K(n)(K(n)^2 - 3K(-n))$	7.4
Argand	IV.15	vertices of the inscribed hexagon	6.2
Brahmagupta–Fibonacci		composition of plane rotations; $F_n^2 + F_{n+1}^2 = F_{2n+1}$; $y^2 + z^2 = 4PQ$	5.3, 7.2, 8
Euler four-square		composition of space rotations; Pythagorean quadruples	8
Sophie Germain		difference of squares twice over the Gaussian integers at $e^{i\pi/4}$	6.1
Sum of odd numbers		accumulation of intervals into nodes; cubes as runs (Nicomachus)	3.2, 4
Extreme and mean ratio	II.11, VI.30	the Fibonacci equation $y^2 - y - 1 = 0$	7.1

11 Conclusion

The notable identities of the catalogue each carry a step of the constructions examined here. The difference of squares of Euclid II.5 carries the node intervals, the boundaries of the Geometric Matrix, the Smith chart ratio, the cubes on triangular nodes and the closure of the space diagonal. Legendre's identity builds the rotation triple, and the Pythagorean nodes $n = 4T(\ell)$ are the triples of Pythagoras's rule. The two-square identity of Brahmagupta and Fibonacci composes ro-

tations in the plane, and Euler's four-square identity composes them in space, where the quadruple $(3, 4, 12, 13)$ closes the diagonal over the 3-4-5 triangle. The Sophie Germain identity is the difference of squares taken twice at $e^{i\pi/4}$, and the difference of cubes with Argand's identity splits the vertices of the inscribed hexagon, whose perimeter ratio 3 and the ratio $22/7$ are the first two convergents of π. Consecutive Fibonacci numbers give the rotation triples $(F_{n-1}F_{n+2},\ 2F_nF_{n+1},\ F_{2n+1})$, and the binomial powers, Cauchy's cube, Stevin's identity and its three-factor extension, Gauss's identity and the trinomial square and cube carry the power relations of the Lucas numbers and the Tribonacci power sums. The quadratic identities are propositions II.4 to II.10 of Euclid's Book II, and the Fibonacci equation is the reciprocal of the cut of II.11. The norm identities belong to the split-complex numbers, the Gaussian integers, the Eisenstein integers, the integers of $\mathbb{Q}(\sqrt{5})$ and the quaternions, the Tribonacci roots to a cubic field, and the remaining identities are the polynomial expansions that act on these conjugates. These identities and their unification are classical. The map that places them at the steps of the constructions is what this paper contributes. Each identity of the school list stands beside its classical source and its later names, so the same relation can be recognized from Euclid's cut line to the conjugate, the norm and the rotation. The sequence from the root and the square (Section 3) to the triple (Section 5) and the quadruple (Section 8) has a classical identity at each step.

References

[1] C. Johnson (2026). Fermions and Bosons as Geometric Complementarity: Quadratic Coordinates, Rotational Classes, and State Multiplicity. SemanticDrift.

[2] C. Johnson (2026). The Geometric Matrix: Rational Derivation of the Pythagorean Triangle. SemanticDrift.

[3] C. Johnson (2026). From Euclid II.5 to the Smith Chart: A Shared Partition Ratio. SemanticDrift.

[4] C. Johnson (2025). The Fibonacci Principle. SemanticDrift.

[5] C. Johnson (2026). The Taylor-Doppler Identity: Frequency Transport on a Phase Manifold. SemanticShift.

[6] T. L. Heath (1926). The Thirteen Books of Euclid's Elements, 2nd ed. Cambridge University Press. Vol. 1: Book II, Propositions 4 to 11, with commentary; Vol. 2: Book IV, Proposition 15; Book VI, Proposition 30.

[7] Brahmagupta (628). Brāhmasphuṭasiddhānta.

[8] Leonardo Pisano (Fibonacci) (1225). Liber Quadratorum. Translated by L. E. Sigler as The Book of Squares, Academic Press, 1987.

[9] L. Euler (1749). Letter to C. Goldbach, 12 April 1749.

[10] W. R. Hamilton (1844). On quaternions; or on a new system of imaginaries in algebra. Philosophical Magazine, 25, 10–13.

[11] T. Levi-Civita (1920). Sur la régularisation du problème des trois corps. Acta Mathematica, 42, 99–144.

[12] Nicomachus of Gerasa. Introduction to Arithmetic. Translated by M. L. D'Ooge, Macmillan, 1926.

[13] Proclus. A Commentary on the First Book of Euclid's Elements. Translated by G. R. Morrow, Princeton University Press, 1970.

[14] H. Bondi (1964). Relativity and Common Sense. Doubleday.

[15] G. H. Hardy and E. M. Wright (1938). An Introduction to the Theory of Numbers. Oxford University Press.

[16] Archimedes. Measurement of a Circle. In T. L. Heath, The Works of Archimedes, Cambridge University Press, 1897.

[17] J. Binet (1843). Mémoire sur l'intégration des équations linéaires aux différences finies. Comptes Rendus de l'Académie des Sciences, 17, 559–567.

[18] É. Lucas (1878). Théorie des fonctions numériques simplement périodiques. American Journal of Mathematics, 1, 184–240.

[19] G. D. Cassini (1680). Presented to the Académie Royale des Sciences; Histoire de l'Académie Royale des Sciences, Paris, vol. 1 (1733), p. 201.

[20] R. Simson (1753). An explication of an obscure passage in Albert Girard's commentary upon Simon Stevin's works. Philosophical Transactions of the Royal Society, 48, 368–377.

[21] O. Rodrigues (1840). Des lois géométriques qui régissent les déplacements d'un système solide dans l'espace. Journal de Mathématiques Pures et Appliquées, 5, 380–440.

[22] P. Kustaanheimo and E. Stiefel (1965). Perturbation theory of Kepler motion based on spinor regularization. Journal für die reine und angewandte Mathematik, 218, 204–219.

[23] O. Neugebauer (1957). The Exact Sciences in Antiquity, 2nd ed. Brown University Press.

[24] J. Høyrup (2002). Lengths, Widths, Surfaces: A Portrait of Old Babylonian Algebra and Its Kin. Springer.

[25] E. Robson (2001). Neither Sherlock Holmes nor Babylon: a reassessment of Plimpton 322. Historia Mathematica, 28, 167–206.

[26] S. Unguru (1975). On the need to rewrite the history of Greek mathematics. Archive for History of Exact Sciences, 15, 67–114.

[27] J. L. Berggren (2016). Episodes in the Mathematics of Medieval Islam, 2nd ed. Springer.

[28] F. Viète (1615). De aequationum recognitione et emendatione. Paris.

[29] A. Girard (1629). Invention nouvelle en l'algèbre. Amsterdam.

[30] A. de Moivre (1730). Miscellanea Analytica de Seriebus et Quadraturis. London.

[31] T. L. Heath (1910). Diophantus of Alexandria: A Study in the History of Greek Algebra, 2nd ed. Cambridge University Press.

[32] T. Levi-Civita (1906). Sur la résolution qualitative du problème restreint des trois corps. Acta Mathematica, 30, 305–327.

[33] T. Koshy (2001). Fibonacci and Lucas Numbers with Applications. Wiley.

[34] L. E. Dickson (1920). History of the Theory of Numbers, Vol. II: Diophantine Analysis. Carnegie Institution of Washington.

[35] G. Sobczyk (1995). The hyperbolic number plane. The College Mathematics Journal, 26, 268–280.

[36] D. F. Mansfield and N. J. Wildberger (2017). Plimpton 322 is Babylonian exact sexagesimal trigonometry. Historia Mathematica, 44, 395–419.